**BUSINESS/IT DIVISION SYLLABUS**

# Cisco Introduction to Networks
# CNT140AB Section 14435
# Fall 2023

**Instructor:** Wayne Kibbe

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**Class Dates:** Mon. 8/28 – Wed. 11/22
**Days:** Online

**Times:** Online

**Online:**

- I will be holding a few WebEx sessions to go over materials and notes. They will also be recorded.

**Final exam time:** Wed. Nov.22 until 11:59 pm

**Textbook:** Recommended (not required)

- **Title**: Introduction to Networks Companion Guide (CCNAv7)
- **Author**: Cisco Networking Academy
- **Publisher**: Cisco Press
- **ISBN-13**: 978-0-13-663366-2

The class is based on and uses content from two websites that are included in the course.

**Online:**
This course is fully online. To attend an in-person class session and use the lab facilities on campus all students must follow safety protocols and observe social distancing guidelines. For information regarding appropriate face coverings and protocols, please visit our COVID website: www.paradisevalley.edu/covid-information/covid. These policies will remain in effect for the entire semester or until further direction from health officials. Paradise Valley Community College values the delivery of quality education as well as the health and safety of all employees and students. Please stay home if you are feeling ill. If you have been confirmed positive, experiencing COVID-19 symptoms, or believe you have been exposed to COVID-19, you must self-report to Maricopa Community Colleges Risk Management by using the COVID-19 reporting forms here.

**Supplies:**

- Pen or Pencil and Legal Notepad
- USB flash drive, external hard drive, or use Google Drive to save files
  - Student generated data is not stored on classroom computers so saving your data is critical for success.

**Course description and/or overview:**

Focus on the architecture, structure, functions, components, and models of the Internet and other computer networks. Principles and structure of IP addressing and the fundamentals of Ethernet concepts, media, and operations are introduced. Students will build simple LANs, perform basic configurations for routers and switches, and implement IP addressing schemes. Preparation for Cisco certification examination. Prerequisites: None.

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**Course Objectives and Competencies:**

Upon completion of this course, you will have these skills:

- Explain the advances in modern network technologies.
- Implement initial settings including passwords, IP addressing, and default gateway parameters on a network switch and end devices.
- Explain how network protocols enable devices to access local and remote network resources.
- Explain how physical layer protocols, services, and network media support communications across data networks.
- Calculate numbers between decimal, binary, and hexadecimal systems.
- Explain how media access control in the data link layer supports communications across networks.
- Explain how Ethernet operates in a switched network.
- Explain how routers use network layer protocols and services to enable end-to-end connectivity.
- Explain how ARP and ND enable communication on a local area network.
- Implement initial settings on a router and end devices.
- Calculate an IPv4 subnetting scheme to efficiently segment your network.
- Implement an IPv6 addressing scheme.
- Use various tools to test network connectivity.
- Compare the operation of transport layer protocols in supporting end-to-end communication.
- Explain the operation of the application layer in providing support to end-user applications.
- Configure switches and routers with device hardening features to enhance security.
- Troubleshoot connectivity in a small network.

**Instructional Contact Hours & Minimum Expectations for Coursework**
For this 12-week class, students should expect to spend 4 hours/week in class and 12 or more hours/week outside of class. However, hours may vary based on student's own skill levels. Students will be required to complete course work both during and **outside** of class.

**Web Based 3rd Party Software**

In this course, we will use Cisco's netacad.com as a web-based 3rd party tool. This tool will be used to complete course examinations and quizzes along with accessing the course reading materials. Students may be required to submit work and/or download information from these tools. There is, therefore, some risk that individuals electing to use the products and services made available by these tools may place any student information shared with the tool vendor at a risk of disclosure.

**Netacad.com & Packet Tracer**

- https://www.netacad.comLinks to an external site. (Links to an external site.) (Netacad uses the Moodle LMS).

- **Terms of use:**
  - https://www.netacad.com/sites/default/files/terms/cisco_networking_academy_terms_and_conditions_july_2016_v3_english.pdfLinks to an external site. (Links to an external site.)
- **Privacy Policy:**
  - https://www.cisco.com/c/en/us/about/legal/privacy.html (Links to an external site.) (Links to an external site.)
- **Accessibility Statement:**
  - https://www.netacad.com/accessibilityLinks to an external site. (Links to an external site.)

**Wireshark®**

- https://www.wireshark.org/ (Links to an external site.) (Links to an external site.)

- **Terms of Use**:
  - Wireshark is available under the GNU General Public License (Links to an external site.) (Links to an external site.) version 2.

**TeraTerm**

- https://osdn.net/projects/ttssh2/downloads/72009/teraterm-4.105.exe/ (Links to an external site.) (Links to an external site.)
- Tera Term is open source free software terminal emulator supporting UTF-8 protocol, TTSSH supports SSH2 protocol.

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**Grading Standards:**
See below for grading scale and point values

| **Grading Scale:** | | |
|---|---|---|
| | 90 – 100% | A |
| | 80 – 89% | B |
| | 70 – 79% | C |
| | 60 – 69% | D |
| | less than 60% | F |

Weighting Rubric:

| Group | Weight |
|---|---|
| | |
| Weekly Assignments | 30% |
| Discussions | 10% |
| Exams | 20% |
| Skills Based Assessment | 20% |
| Final Exam | 20% |
| **Total** | 100% |

| WEEK | ASSIGNMENT/TEST/POINT VALUE |
|---|---|
| **Week 1**<br>**8/28 – 9/2** | Syllabus Acknowledgement: 5pts<br>Week 1 Assignments: 100pts |
| **Week 2**<br>**9/3 – 9/9** | Read Modules 1-3<br>Week 2 Assignments: 100pts |
| **Week 3**<br>**9/10 – 9/16** | Module 1-3 Exam: 100pts<br>Week 3 Assignments: 100pts<br>Read Modules 4-7 |
| **Week 4**<br>**9/17 – 9/23** | Module 4-7 Exam: 100pts<br>Read Modules 8-10<br>Week 4 Assignments: 100pts |
| **Week 5**<br>**9/24 – 9/30** | Module 8-10 Exam: 100pts<br>Read Module 11-13<br>Week 5 Assignments: 100pts |
| **Week 6**<br>**10/1 – 10/7** | Week 6 Assignments: 100pts |
| **Week 7**<br>**10/8 – 10/14** | Week 7 Assignments: 100pts |
| **Week 8**<br>**10/15 – 10/21** | Module 11-13 Exam: 100pts<br>Read Modules 14-15<br>Week 8 Assignments: 100pts |
| **Week 9**<br>**10/22 – 10/28** | Week 9 Assignments: 100pts |

| | |
|---|---|
| **Week 10**<br>**10/29 – 11/4** | Module 14-15 Exam: 100pts<br>Read Modules 16-17<br>Week 10 Assignments: 100pts |
| **Week 11**<br>**11/5 – 11/11** | Module 16-17 Exam: 100pts<br>Week 11 Assignments: 100pts |
| **Week 12**<br>**11/12 – 11/22** | Skills Based Assessment: 100pts<br>Course Feedback: Pre-requisite for taking Final<br>Final Exam: 100pts |

- **Schedule may change during the semester. Changes will be announced in class and posted in the PVC Canvas Shell.
- **All Students must take the final exam to achieve credit for this course regardless of their grade.**

**Late Work Policy:**

- Late Labs and Packet Tracer activities will automatically receive %50 of the credit earned for the assignment.
  - Please do not copy and paste answers from Instructor's Lab manuals or other resources. This is the easiest and fastest way to get zero (0) points on an assignment.
- Missed exams will receive a grade of zero (0).

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**Classroom Etiquette:**

Cell phones must be turned off during class instruction. Please arrive on time and listen respectfully while the instructor or other class members are speaking. Recording of lectures is not allowed unless you receive special permission from the instructor. Food and drink are not permitted in the classroom.

When on campus, students MUST follow all safety protocols (e.g. wear a mask or face covering, wash hands when entering the classroom, consent to temperature checks, etc.) AND observe social distancing guidelines.

In addition, during remote instruction or while using telecommunications platforms (WebEx, Zoom, Google Meets, etc.), remember to:

- Present yourself professionally at all times, just like you would in a classroom.
- Conduct yourself in all chat rooms and on discussion posts with proper decorum.
- Assure the workspace is free of excess noise.
- Mute your microphone when you are not speaking.
- Ensure your background is suitable, especially when the camera is on.
- Be on time, and pay attention to the presenter.
- Maintain the privacy of any class recordings. In other words, no sharing beyond the classroom learning platform.
- Always monitor video, microphone, and screen sharing settings to ensure only appropriate content is broadcast through the platform.

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**Classroom Conduct Policy:**
You are expected to treat your instructor and your classmates with respect. In all correspondence, whether communicating in person, virtually or online, you should show respect for the viewpoints of others who may disagree with you or see things from a different perspective. Criticizing, ridiculing, insulting, or belittling others will not be accepted. Avoid coarse, rough, or rude language and sarcasm. Keep in mind that electronic communications do not have the advantage of nonverbal cues that are so much a part of interpersonal communication. Humor or satire can sometimes be misinterpreted in strictly electronic communication forums. Observe good grammar and spelling.

If the instructor determines that the student's conduct is not consistent with the classroom conduct policy, the student may be removed from the discussion forum, virtual lecture or classroom, or dropped from the course entirely.

**For Your Protection:**
It is your responsibility to SAVE your assignments in a folder, on your personal computer or another removable media. Keep these copies in a safe place until you have received your final grade.

**Withdrawal Policy:**
Business/IT Division recommended policy state that instructors **may** withdraw students who accumulate unofficial absences in excess of the number of times a class meets per week. On-line students who do not participate (turn in assignments) will be withdrawn. Students may withdraw themselves without an instructor's signature, see date in my.maricopa.edu. If a student requests a withdrawal after the self- withdraw period, the instructor may withdraw a student with a Y if the cumulative grade is failing at that point.

**Attendance:**
Only persons who are registered for the class may attend that class. Attendance requirements are determined by the course instructor. Students who do not meet the attendance requirement, as determined by the course instructor, may be withdrawn.

Students who fail to attend the first scheduled class meeting, or to contact the instructor regarding absence before the first scheduled class meeting may at the option of the instructor be withdrawn.

Online students who do not post in Canvas the first week of class risk being withdrawn from the class. Posting the first week in Canvas the first week is equivalent to showing up on the first day of class.

At the beginning of each course, each faculty member will provide students with written attendance requirements. It is the student's responsibility to

consult with the instructor regarding official or unofficial absences.

Students bear the responsibility of notifying the Office of Admissions and Records when they discontinue studies in a course or at the college. Refer to the Withdrawal Procedures.

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**Student Maricopa E-mail policy:**

**Your MEID@maricopa.edu e-mail account is the official form of communication between PVCC and you. You are responsible for reading college messages and responding accordingly. You should check your my.maricopa.edu account for official college/district messages and information regarding tuition and fees.**

**Official Absences -- MCCCD policy states that:**
Official absences are those that occur when students are involved in an official activity of the college, i.e., field trips, tournaments, athletic events, and present an official absence excuse form. Absences for such events shall not count against the number of absences allowed by an instructor or department. Students who must miss a class for an official reason must obtain an official absence verification card from the appropriate dean or associate dean and present it to the appropriate instructor(s) before the absence. Prior arrangements must be made with each instructor for make-up work. If prior arrangements have been made, the students will not be penalized.

Other official absences include jury duty and subpoenas. Appropriate documentation will be required. Prior arrangements must be made with each instructor for makeup work. If prior arrangements have been made, the students will not be penalized.

In the event of the death of an immediate family member, absences for periods of up to one week will not be counted against the number of absences allowed by an instructor or department. Students should contact instructor(s) as soon as possible to arrange for make-up work. Appropriate documentation will be required (for example, a copy of the obituary or funeral program). In specialized programs that require clinical rotations, this regulation may not apply.

**Religious Holidays -- MCCCD policy states that:**
Students shall have the right to observe major religious holidays without penalty or reprisal by any administrator, faculty member or employee of the Maricopa Community Colleges. Absences for such holidays shall not count against the number of absences allowed by an instructor or department. At least one week before the holiday, students shall submit to their instructor(s) a written statement that includes both the date of the holiday and the reason why class attendance is impossible. Prior arrangements must be made with each instructor for make-up work. If prior arrangements have been made, the student will not be penalized.

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**Title IX Statement:**
**Addressing Incidents of Title IX Sexual Harassment**

In accordance with Title IX of the Education Amendments of 1972, the MCCCD prohibits unlawful sexual harassment against any participant in its education programs or activities. Sexual harassment includes quid pro quo (this for that) harassment, hostile environment, sexual assault, dating/domestic violence, and stalking. This prohibition against sexual harassment - including sexual violence - applies to students, MCCCD employees, and visitors to campus.

The policy of the MCCCD is to provide an educational, employment, and business environment free of sexual violence, unwelcome sexual advances, requests for sexual favors, and other verbal or physical conduct or communications constituting sexual harassment as prohibited by state and federal law. Incidents of Sexual Misconduct should be reported to the college Title IX Coordinator, as outlined in policy. MCCCD will provide on its Title IX and Preventing Sexual Harassment webpage a link to all Title IX Coordinators at the MCCCD. Reports may be filed anonymously at: https://district.maricopa.edu/consumer-information/reporting.

**Student Disabilities:**
**Classroom Accommodations for Students with Disabilities**

In accordance with the Americans with Disabilities Act (ADA), the Maricopa County Community College District (MCCCD) and its associated colleges are committed to providing equitable access to learning opportunities to students with documented disabilities (e.g. mental health, attentional, learning, chronic health, sensory, or physical). Each class/term/semester that a student needs academic adjustments/accommodations, the qualified student is required to work with the Disability Resources and Services Office (DRS) at their individual college(s). College DRS offices may only offer accommodations/academic adjustments for their own college. Contact with the DRS should be made as soon as possible to ensure academic needs are met in a reasonable time. New and returning students must request accommodations each semester through DRS Connect online services. To learn more about this online service process, please contact your local DRS office.

If you have not yet established services through DRS, but have a temporary health condition or permanent disability that requires accommodations, you are welcome to contact DRS by using the information listed on the following webpage: https://district.maricopa.edu/consumer-information/disability-resources/contacts. The DRS office offers resources and coordinates reasonable accommodations for students with disabilities and/or temporary health conditions qualifying for accommodations/academic adjustments. Reasonable accommodations are established through an interactive process

between you, your faculty, and DRS; and only those academic adjustments/reasonable accommodations granted by the DRS are recognized by the college and MCCCD. It is the policy and practice of the MCCCD to create inclusive and accessible learning environments consistent with federal and state law.

**Students Rights and Responsibilities:**

Students are responsible to read and understand the District wide Scholastic Standards found in the current Student Handbook and are responsible for the college policies included in the college catalog.

**STUDENT CODE OF CONDUCT**

Students engaging in the following are subject to disciplinary sanctions outlined in the PVCC Catalog, Student Policies Section.

Acts of Dishonesty examples include:

- Plagiarism
- Furnishing false information
- Falsifying records related to coursework
- Forgery, alteration, misuse of any college document
- Tampering with the election of any college-recognized official
- Disruption or obstruction of teaching, research, administration, disciplinary proceedings, or college activities. An instructor can remove a student from class for disciplinary reasons.
- Physical abuse, verbal abuse, threats, intimidation, harassment, coercion and/or conduct that threatens the health or safety of any person.
- Attempted or actual theft
- Failure to comply with direction of college officials or law enforcement officers
- Unauthorized possession, duplication or use of keys to any college premises
- Violation of federal, state, or local laws on college campus or at college sponsored activities
- Use, possession or distribution of narcotics or other controlled substances
- Use, possession, or distribution of alcoholic beverages, or public intoxication
- Illegal or unauthorized possession of firearms, explosives, weapons, or dangerous chemicals
- Participation in a college demonstration that disrupts the normal operations of the college
- Obstruction of the free flow of pedestrian or vehicular traffic on college premises
- Conduct which is disorderly, lewd or indecent
- Theft or other abuse of computer time
- Theft or other copying of proprietary software found on PVCC computers and network devices
- Abuse of the judicial system

**Policy on Cheating:**

Students caught cheating will receive a 0 for the assignment, or an F for the semester. You will also be referred to the Dean of Academic Affairs for probation or dismissal from the college.

Students will be notified by the instructor of any changes in course requirements or policies.

Students are responsible for the information contained in this syllabus.

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**SYLLABUS ACKNOWLEDGMENT RECEIPT**

NAME:

_____

Last First Middle

CLASS:

_____

Course Prefix Section Title

EMAIL ADDRESS:

_____

MAJOR:

_____

SEMESTER:_____TODAY'S DATE:_____

The instructor has given me a copy of the course syllabus for this class. I have read the syllabus and understand and accept responsibility for all information covered in the syllabus. I, furthermore, accept responsibility for the college policies included in the college catalog and student handbook.

_____

Student Signature